

|                                                |                                            |
|------------------------------------------------|--------------------------------------------|
| Program : <b>Diploma in Polymer Technology</b> |                                            |
| Course Code : <b>6091B</b>                     | Course Title: <b>Fibres And Composites</b> |
| Semester : <b>6</b>                            | Credits: <b>4</b>                          |
| Course Category: <b>Program Elective</b>       |                                            |
| Periods per week: <b>4 (L:3, T:1, P:0)</b>     | Periods per semester: <b>60</b>            |

### Course Objectives:

- To acquire the knowledge of the production, general characteristics and classification of natural and synthetic fibres.
- To understand the polymer composites and its constituents along with processing techniques of composite products.

### Course Prerequisites:

| Topic                                                                       | Course code | Course name                     | Semester |
|-----------------------------------------------------------------------------|-------------|---------------------------------|----------|
| General idea about different type of polymerization chemistry and technique |             | Fundamentals of polymer science | 2        |
| Different plastic materials                                                 |             | Plastic Technology              | 4        |

### Course Outcomes:

On completion of the course, the student will be able to:

| CO <sub>n</sub> | Description                                                                                       | Duration (Hours) | Cognitive level |
|-----------------|---------------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | Explain the manufacture, properties and applications various natural fibres                       | 13               | Understanding   |
| CO2             | Describe the manufacture, properties and applications various semi-synthetic and synthetic fibres | 15               | Understanding   |
| CO3             | Explain the constituents and classification of composite                                          | 15               | Understanding   |
| CO4             | Paraphrasing the processing methods of different composite products                               | 15               | Understanding   |
|                 | Series Test                                                                                       | 2                |                 |

**CO – PO Mapping:**

| <b>Course Outcomes</b> | <b>PO1</b> | <b>PO2</b> | <b>PO3</b> | <b>PO4</b> | <b>PO5</b> | <b>PO6</b> | <b>PO7</b> |
|------------------------|------------|------------|------------|------------|------------|------------|------------|
| <b>CO1</b>             | 3          |            |            |            |            |            |            |
| <b>CO2</b>             | 3          |            |            |            |            |            |            |
| <b>CO3</b>             | 3          |            | 3          |            |            |            |            |
| <b>CO4</b>             | 3          |            | 3          |            |            |            |            |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

**Course Outline:**

| <b>Module Outcomes</b>                                                                                                                                                                                                                                 | <b>Description</b>                                                                                                       | <b>Duration (Hours)</b> | <b>Cognitive Level</b> |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|-------------------------|------------------------|
| <b>CO1</b>                                                                                                                                                                                                                                             | <b>Explain the manufacture, properties and applications various natural fibres</b>                                       |                         |                        |
| M1.01                                                                                                                                                                                                                                                  | State the definition of fibre, yarn, cord, strand, tex, denier, tenacity, warp, weft and twist                           | 3                       | Understanding          |
| M1.02                                                                                                                                                                                                                                                  | Identify the different natural fibres and classify them according to their origin                                        | 3                       | Understanding          |
| M1.03                                                                                                                                                                                                                                                  | Explain the source, processing properties and applications of fibres such as Cotton, Jute, Coir, Silk.                   | 7                       | Understanding          |
| <b>Contents:</b><br>Define–fibre, yarn, cord, strand, tex, denier, tenacity, warp, weft and twist. Natural fibres - classification, vegetable, animal and mineral fibres - examples. Processing - properties - applications - Cotton, Jute, Coir, Silk |                                                                                                                          |                         |                        |
| <b>CO2</b>                                                                                                                                                                                                                                             | <b>Describe the manufacture, properties and applications various semi-synthetic and synthetic fibres</b>                 |                         |                        |
| M2.01                                                                                                                                                                                                                                                  | Explain the different spinning processes such as dry spinning and wet spinning for the conversion of polymer into fibres | 3                       | Understanding          |
| M2.02                                                                                                                                                                                                                                                  | Outline the manufacture, properties and Applications of Viscose Rayon,                                                   | 4                       | Understanding          |
| M2.03                                                                                                                                                                                                                                                  | Explain the raw polymer, Fiber formation, properties and applications of Nylons, Polyester, and Aramid                   | 5                       | Understanding          |

|                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                             |   |               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|---|---------------|
| M2.04                                                                                                                                                                                                                                                                                                                                                                                                                        | Generalize manufacture, Properties, Applications of different forms-grades-characteristics of glass fibres and carbon fibre | 3 | Understanding |
|                                                                                                                                                                                                                                                                                                                                                                                                                              | Series Test – I                                                                                                             | 1 |               |
| <b>Contents:</b><br>Synthetic Fibres - Spinning process-Melt spinning, Solution spinning. Manufacture, Properties and Applications - Viscose Rayon. Raw polymer, Fiber formation, properties and applications - Nylons, Polyester, and Aramids. Glass fibre Manufacture, Properties, applications, different forms - grades - characteristics. Carbon fibre - Manufacture, properties, applications                          |                                                                                                                             |   |               |
| <b>CO3</b>                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Explain the constituents and classification of composite</b>                                                             |   |               |
| M3.01                                                                                                                                                                                                                                                                                                                                                                                                                        | Describe polymer composite and its classifications                                                                          | 3 | Understanding |
| M3.02                                                                                                                                                                                                                                                                                                                                                                                                                        | Explain the function and performance characteristics of constituents of composites                                          | 6 | Understanding |
| M3.03                                                                                                                                                                                                                                                                                                                                                                                                                        | Generalize design factors for composite products                                                                            | 3 | Understanding |
| M3.04                                                                                                                                                                                                                                                                                                                                                                                                                        | Compare composites with conventional materials                                                                              | 3 | Understanding |
| <b>Contents:</b><br>Polymer composites-Classification-Polymer matrix and reinforcement. Composite constituents - Function, performance characteristic - Resins - UP Resin, epoxy Reinforcements - glass, coir, Catalyst, Accelerators and Fillers - Examples. Comparison with conventional materials - advantages and disadvantages. Composites - Rule of Mixtures. Design factors - process, section thickness and corners. |                                                                                                                             |   |               |
| <b>CO4</b>                                                                                                                                                                                                                                                                                                                                                                                                                   | <b>Paraphrasing the processing methods of different composite products</b>                                                  |   |               |
| M4.01                                                                                                                                                                                                                                                                                                                                                                                                                        | Explain the various open mould process for composite products                                                               | 2 | Understanding |
| M4.02                                                                                                                                                                                                                                                                                                                                                                                                                        | Explain the various closed mould process for composite products                                                             | 2 | Understanding |
| M4.03                                                                                                                                                                                                                                                                                                                                                                                                                        | Generalize the materials for mould making for composite products                                                            | 3 | Understanding |
| M4.04                                                                                                                                                                                                                                                                                                                                                                                                                        | Describe the manufacturing of different composite products                                                                  | 8 | Understanding |
|                                                                                                                                                                                                                                                                                                                                                                                                                              | Series Test – II                                                                                                            | 1 | Understanding |

**Contents:**

Composite products - Processing techniques open mould - Hand layup, Spray layup, Vacuum bag Moulding, Filament winding Pressure bag moulding, closed mold - Pultrusion, centrifugal casting, and Reaction injection moulding, Compression molding - Differentiate - SMC, DMC. Moulds - FRP, POP, wood - advantages, disadvantages. Manufacture method- Boat hull, Roofing sheet, tanks, Pipes. Define - Hybrid composite and nano composite- advantages and applications

**Text / Reference:**

| T/R | Book Title/Author                                                                        |
|-----|------------------------------------------------------------------------------------------|
| T1  | Fibre science and technology - Premamoy Ghosh(Publisher Tata McGraw-Hill Education 2004) |
| R1  | Man made fibres -R.W Moncrieff (Publisher –Iliffe 4th Edition)                           |
| R2  | FRP technology - R.G Weatherhead (Applied Science Publishers 1980)                       |

**Online Resources:**

| Sl.No | Website Link                                                                |
|-------|-----------------------------------------------------------------------------|
| 1     | <a href="https://link.springer.com">https://link.springer.com</a>           |
| 2     | <a href="https://www.frontiersin.org">https://www.frontiersin.org</a>       |
| 3     | <a href="https://www.irjet.net/">https://www.irjet.net/</a>                 |
| 4     | <a href="https://www.sciencedirect.com/">https://www.sciencedirect.com/</a> |